SEAL: Curriculum Transfer Gap in Self-Editing Language Models

1 · Data & Verification

Fact Pool

`data/fact_pool.json`

19 verified post-2024 facts

tech · science · politics · sports
passage + title + test QA pairs

Novelty Verification

`verify_novelty.py`

Qwen2.5-1.5B-Instruct scores
0% prior knowledge

2 · Curriculum Selection

Curriculum Strategy

PlannedCurriculum (Human)

Fixed static order · used in C3

SelfGeneratedCurriculum (Model)

LLM picks next fact + logs reasoning
used in C2, C4, C5

3 · Self-Edit Loop (19 iterations)

Synthetic Data Gen

`generate_synthetic_data`

4 QA pairs from passage

SFT Adaptation Method

C2 · Vanilla LoRA

Standard LoRA SFT, no constraint

C3/C4 · O-LoRA

`src/orthogonality.py · λ=0.1`

$L_{SFT} + \lambda \Sigma ||A_t^T A_i||^2_F$

C5 · LoRI

A frozen random projection

top-k mask on B (80% sparse)

Cumulative QA Evaluation

Greedy decode over

all facts 0 ... t